

ARMS DOWN DEEP SQUAT



Test Objective for the Arms Down Deep Squat:

The arms down deep squat assessment tests for bilateral symmetrical mobility of the hips, knees and ankles.

How to Perform the Arms Down Deep Squat:

The patient assumes the starting position by placing their feet together and arms out in front of their body. The feet should be in the sagittal plane - toes pointing straight ahead. Have the patient slowly descend as deeply as possible into a squat position, allowing their arms to remain out front. The squat position should be assumed with the heels on the floor and head and chest facing forward. If they break parallel (hips below knees) at the bottom of the squat, now have them lower their arms to touch their fists on floor within their footprint. They should maintain sagittal plane throughout movement.

Functional Criteria:

- Thighs break parallel
- Touches fists to floor within footprint
- Maintains Sagittal plane
- Without excessive effort and/or motor control

Common Compensations:

- Ankles externally rotate
- Heels lift off ground
- Falls over

Tips for Testing:

- Observe from the front and side.
- Foot position should remain unchanged throughout the movement.
- Knees are allowed to move outward during the squat.
- Do not coach the movement; simply repeat the instructions if needed.
- Was there pain?

Anterior View:



Side View:



ARMS DOWN DEEP SQUAT BREAKOUT



Arms Down Deep Squat Breakout Rationale:

Ankle Mobility Problem

One of the most common limitations to prevent functional squatting is limited ankle mobility. We begin the Arms Down Deep Squat breakout by looking at ankle mobility - this includes ankle dorsiflexion, inversion, and eversion.

To assess closed chain ankle dorsiflexion (knee flexed), we use the Active Tandem Ankle Dorsiflexion - Knee Flexed Test. If closed chain ankle dorsiflexion is functional and non-painful, consider ankle dorsiflexion with the knee flexed normal and continue forward to assess ankle inversion/eversion with the Active Seated Ankle Inversion/Eversion Test. If Active Tandem Ankle Dorsiflexion - Knee Flexed Test is dysfunctional and/or painful, proceed to the Passive Prone Ankle Dorsiflexion - Knee Flexed Test to differentiate between a mobility or stability/motor control dysfunction.

If Passive Prone Ankle Dorsiflexion - Knee Flexed Test is dysfunctional and/or non-painful, there is a dorsiflexion mobility dysfunction. If the test is now functional and non-painful, there is an ankle dorsiflexion SMCD. If passive dorsiflexion is painful, note where the pain is and come back and examine the region in more detail after completing the entire SFMA. Regardless of the ankle dorsiflexion finding, proceed to assess ankle inversion/eversion.

If the Active Seated Ankle Inversion/Eversion Test is functional and non-painful, consider ankle inversion/eversion normal and continue forward to assess hip mobility. If ankle inversion/eversion is dysfunctional and/or painful, proceed to the Passive Ankle Inversion/Eversion Test to differentiate between a mobility or stability/motor control dysfunction.

A functional and non-painful Passive Ankle Inversion/Eversion Test identifies an inversion/eversion SMCD. If the passive testing is dysfunctional and non-painful, there is an ankle inversion/eversion mobility dysfunction. If Passive Inversion/Eversion test is painful, note where the pain is and come back and examine the region in more detail after completing the entire SFMA.

No matter your findings for the ankle motions, we need to proceed and assess hip and knee mobility.

Hip Flexion and Knee Flexion Problems

We can change the stability requirements for both hip and knee flexion using the Supine Knees to Chest Holding Shins Test. Make sure the patient keeps their knees together and does not flex their lumbar spine. This test allows us to look at both hip flexion and knee flexion at the same time. If the Supine Knees to Chest Holding Shins Test is functional and non-painful, consider hip flexion and knee flexion normal and continue forward to assess internal/external hip rotation with the Active Seated Internal/External Hip Rotation Tests.

If the Supine Knees to Chest Holding Shins Test is dysfunctional and/or painful, we need to remove a part from this test. The easiest way to do this is to eliminate knee flexion by now having the patient hold on to the back of their thighs rather than their shins and repeat the test. Knee

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flexion is no longer challenged in this position.

If the Supine Knees to Chest Holding Thighs Test is now functional and non-painful, there is a knee flexion mobility dysfunction. If the patient still cannot show full hip flexion, there is a hip flexion mobility dysfunction -this does not rule out knee involvement, it just rules in the hip.

If the Supine Knees to Chest Holding Thighs Test is painful, note where the pain is and come back and examine the region in more detail after completing the entire SFMA.

No matter your findings for the hip and knee flexion, we need to proceed and assess internal/external hip rotation.

Hip Rotation Problems

In addition to needing hip flexion to squat, we also need hip rotation. Because there are changes to tissue tension and muscle function for hip rotation when the hip is extended versus flexed, we must now consider the squat requires hip rotation while the hip is also flexed. Even if you have previously assessed hip rotation in the Multi-Segmental Rotation breakout, it is necessary to assess it now again with the hip flexed using the seated position.

We will look at both hip internal and hip external rotation in the seated position - this position allows us to change the stability requirements for the hip. Let's start with internal hip rotation.

If the Active Seated Internal Hip Rotation Test is functional and non-painful, consider hip internal rotation normal in the hip flexed position. Proceed to the Seated External Hip Rotation Test.

If the Active Seated Internal Hip Rotation Test is dysfunctional and/or painful, proceed to the Passive Seated Internal Hip Rotation Test to differentiate between a mobility or stability/motor control dysfunction. If the passive test is functional and non-painful, there is an internal hip rotation SMCD with the hip flexed. If the passive test is dysfunctional and non-painful, there is an internal hip rotation mobility dysfunction. If the Passive Seated Internal Hip Rotation Test is painful, note where the pain is and come back and examine the region in more detail after completing the entire SFMA.

Now let's look at external hip rotation. If the Active Seated External Hip Rotation Test is functional and non-painful, consider hip external rotation normal in the hip flexed position.

If the Active Seated External Hip Rotation Test is dysfunctional and/or painful, proceed to the Passive Seated External Hip Rotation Test to differentiate between a mobility or stability/motor control dysfunction. If the passive test is functional and non-painful, there is an external hip rotation SMCD with the hip flexed. If the passive test is dysfunctional and non-painful, there is an external hip rotation mobility dysfunction. If the Passive Seated External Hip Rotation Test is painful, note where the pain is and come back and examine the region in more detail after completing the entire SFMA.

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Selective Breakout Tests for Arms Down Deep Squat Breakout:



Active Tandem Ankle
Dorsiflexion Test



Passive Prone Ankle
Dorsiflexion Test -
Knee Flexed



Active Seated Ankle
Inversion/Eversion
Test



Passive Ankle
Inversion/Eversion
Test



Supine Knees to Chest
Holding Shins Test



Supine Knees to Chest
Holding Thighs Test



Active Seated
External Hip
Rotation Test



Passive Seated
External Hip
Rotation Test

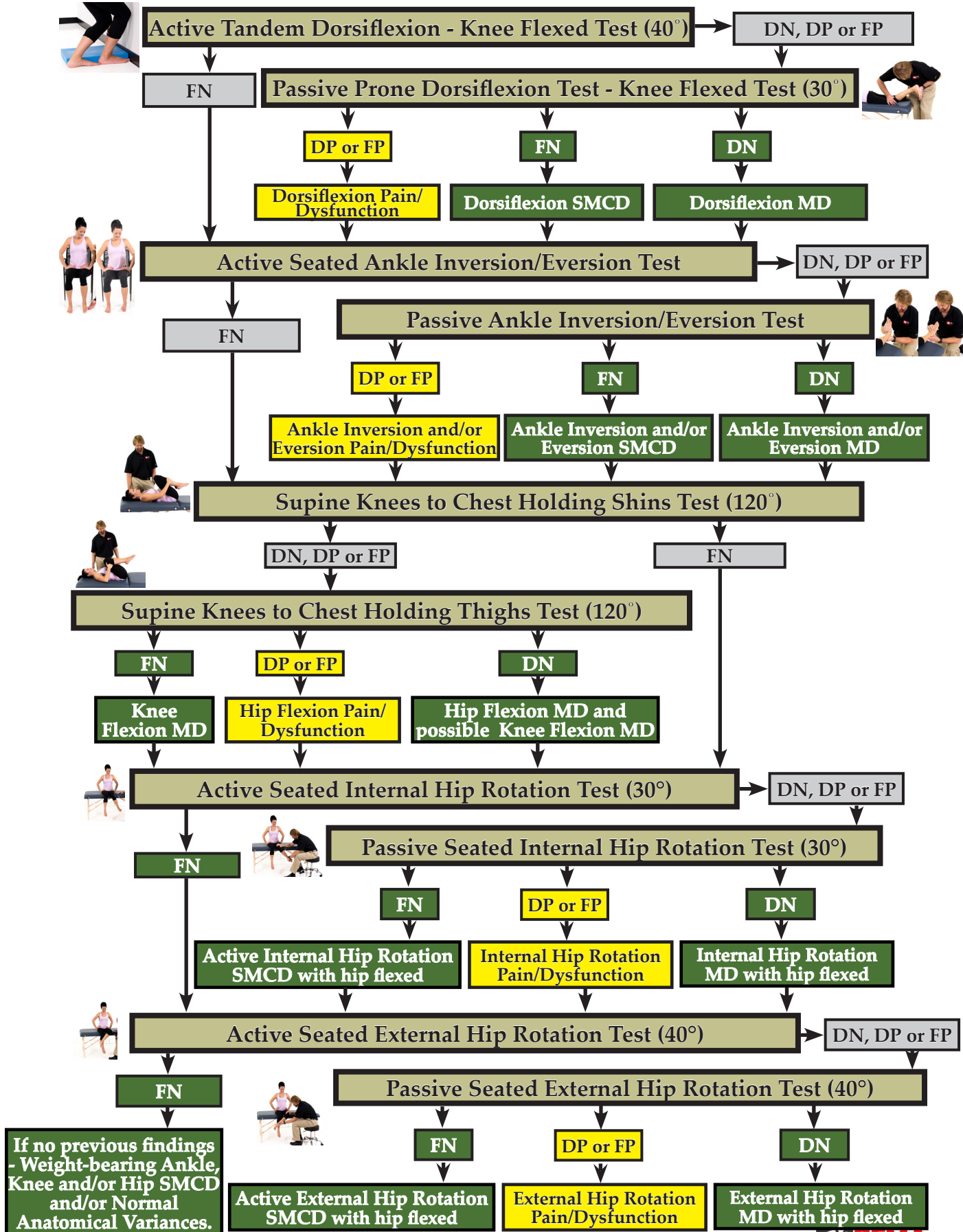


Active Seated
Internal Hip
Rotation Test



Passive Seated
Internal Hip
Rotation Test

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Test Objective for the Active Tandem Ankle Dorsiflexion - Knee Flexed:

To test closed chain ankle dorsiflexion.

How to Perform the Active Tandem Ankle Dorsiflexion - Knee Flexed:

Instruct the patient to assume the tandem gait stance using the wall for support. Tandem stance set-up is the left foot toes against the wall and the right foot toes against left heel, the 2nd/3rd toe in line with left heel. The elbows should be at 90 degrees and in line with their rib cage. Once in position, test knee flexed dorsiflexion of the right ankle by bending knee forward without the heel coming off the ground or valgus collapse. Switch legs for set-up and then repeat with left ankle.

Functional Criteria:

Normal motion for closed chain knee flexed dorsiflexion is 40 degrees.

Additional Information:

Watch carefully for valgus collapse and/or the heel lifting off the ground.

An Airex can be used to keep the feet in proper position - align it along the medial border of front foot and lateral border of back foot.

Possible Findings:

- FN: 40 degrees or greater of closed chain ankle dorsiflexion, no pain
 - » Consider ankle dorsiflexion normal. Proceed to the Active Seated Inversion/Eversion Test.
- FP: 40 degrees or greater of closed chain ankle dorsiflexion, pain
 - » Proceed to Passive Prone Ankle Dorsiflexion Test - Knee Flexed.
- DN: Less than 40 degrees of closed chain ankle dorsiflexion, no pain
 - » Proceed to Passive Prone Ankle Dorsiflexion Test - Knee Flexed.
- DP: Less than 40 degrees of closed chain ankle dorsiflexion, pain
 - » Proceed to Passive Prone Ankle Dorsiflexion Test - Knee Flexed.



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Test Objective for the Passive Prone Ankle Dorsiflexion Test - Knee Flexed:

To test passive ankle dorsiflexion with knee flexed in an unloaded posture to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Prone Ankle Dorsiflexion Test - Knee Flexed:

Instruct the patient to lie prone and flex their knee to 45 degrees. Measure the total passive dorsiflexion of the ankle.

Functional Criteria:

Normal open chain ankle dorsiflexion with knee flexed is 30 degrees.

Additional Information:

Before taking the measurement, make sure the ankle starts in a neutral position.

Possible Findings:

- FN: 30 degrees of passive ankle dorsiflexion with knee flexed, no pain
 - » Indicates a dorsiflexion stability/motor control dysfunction. Proceed to the Active Seated Ankle Inversion/Eversion Test.
- FP: 30 degrees of passive ankle dorsiflexion with knee flexed, pain
 - » Indicates pain with ankle dorsiflexion - complete local biomechanical exam. Proceed to the Active Seated Ankle Inversion/Eversion Test.
- DN: Less than 30 degrees of passive ankle dorsiflexion with knee flexed, no pain
 - » Indicates an ankle dorsiflexion mobility dysfunction - complete local biomechanical exam. Proceed to the Active Seated Ankle Inversion/Eversion Test.
- DP: Less than 30 degrees of passive ankle dorsiflexion with knee flexed, pain
 - » Indicates a painful ankle dorsiflexion mobility dysfunction - complete local biomechanical exam. Proceed to Active Seated Ankle Inversion/Eversion Test.



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Test Objective for the Active Seated Ankle Inversion/Eversion Test:

To test active ankle inversion (supination) and eversion (pronation).

How to Perform the Active Seated Ankle Inversion/Eversion Test:

Instruct the patient to sit on a chair with the knees pelvic-width apart, feet flat on the ground, toes pointing forward and the hands on hips. Then, instruct the patient to invert and evert the feet. Have the person move back and forth between inversion and eversion.

Functional Criteria:

Normal ankle inversion/eversion is the ability to complete the motions. This is a quality test, not quantity - can they do it, or can't they.

Additional Information:

Because of ankle-joint limitations, many people create this motion at the hips. The person should be able to do this without moving the hips or knees.

Possible Findings:

- FN: Ability to actively invert/evert the ankle, no pain
 - » Consider ankle inversion/eversion normal. Proceed to the Supine Knees to Chest Holding Shins Test.
- FP: Ability to actively invert/evert the ankle, pain
 - » Proceed to the Passive Seated Ankle Inversion/Eversion Test.
- DN: Inability to actively invert and/or evert the ankle, no pain
 - » Proceed to the Passive Seated Ankle Inversion/Eversion Test.
- DP: Inability to invert and/or evert the ankle, pain
 - » Proceed to the Passive Seated Ankle Inversion/Eversion Test.



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Test Objective for the Passive Ankle Inversion/Eversion Test:

To test passive inversion (supination) and eversion (pronation) to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Ankle Inversion/Eversion Test:

Instruct the patient to lie supine on table. Stabilize the patient's lower limb and passively invert and evert their feet.

Functional Criteria:

Normal ankle inversion/eversion is the ability to complete the motions. This is a quality test, not quantity - can you do it, or can't you do it.

Additional Information:

You only need to test the directions that were dysfunctional or painful in the Active Seated Inversion/Eversion Test.

Possible Findings:

- FN: Ability to passively invert and/or evert the ankle, no pain
 - » Indicates an ankle inversion and/or eversion (depending on the motion you are testing) stability/motor control dysfunction. Proceed to the Supine Knees to Chest Holding Shins Test.
- FP: Ability to passively invert and/or evert the ankle, pain
 - » Indicates pain with ankle inversion and/or eversion (depending on the motion you are testing) - perform local biomechanical exam. Proceed to the Supine Knees to Chest Holding Shins Test.
- DN: Inability to passively invert and/or evert the ankle, no pain
 - » Indicates an ankle inversion and/or eversion (depending on the motion you are testing) mobility dysfunction - perform local biomechanical exam. Proceed to the Supine Knees to Chest Holding Shins Test.
- DP: An inability to passively invert and/or evert the ankle, pain
 - » Indicates a painful ankle inversion and/or eversion (depending on the motion you are testing) mobility dysfunction - perform local biomechanical exam. Proceed to the Supine Knees to Chest Holding Shins Test.



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Test Objective for the Supine Knees to Chest Holding Shins Test:

To test the mobility of hip flexion and knee flexion in an unloaded posture.

How to Perform the Supine Knees to Chest Holding Shins Test:

Have the patient lie supine and bring both knees up, keeping the knees together. The patient should then grab on top of mid shin and pull the thighs to the chest.

Functional Criteria:

Normal hip flexion with knees bent is 120 degrees with the sacrum remaining in contact with floor/table. This is commonly when the thighs touch to the lower rib cage and the calves to hamstrings.

Additional Information:

Knees should remain together for this test. Look for symmetry between right/left.

Possible Findings:

- FN: 120 degrees of hip flexion with full knee flexion, no pain
 - » Consider hip flexion and knee flexion normal. Proceed to the Active Seated Hip Rotation Tests.
- FP: 120 degrees of hip flexion with full knee flexion, pain
 - » Proceed to the Supine Knees to Chest Holding Thighs Test.
- DN: Less than 120 degrees of hip flexion and/or full knee flexion, no pain
 - » Proceed to the Supine Knees to Chest Holding Thighs Test.
- DP: Less than 120 degrees of hip flexion and/or full knee flexion, pain
 - » Proceed to the Supine Knees to Chest Holding Thighs Test.



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Test Objective for the Supine Knees to Chest Holding Thighs Test:

To test hip flexion to determine if there is mobility dysfunction in the hips or if there is a knee mobility dysfunction. Only perform this test if the patient could not perform the test correctly when holding the shins.

How to Perform the Supine Knees to Chest Holding Thighs Test:

Have the patient lie supine and bring both knees up, keeping the knees together. The patient should then grab the back of the thighs and pull the thighs to the chest.

Functional Criteria:

Normal hip flexion with knees bent is 120 degrees with the sacrum remaining in contact with floor/table. This is commonly when the thighs lay against the lower rib cage.

Additional Information:

This move will help differentiate between a hip flexion mobility dysfunction and a knee flexion mobility dysfunction.

Possible Findings:

- FN: 120 degrees of hip flexion, no pain
 - » Indicates a knee flexion mobility dysfunction - complete local biomechanical exam. Proceed to the Active Seated Hip Rotation Tests.
- FP: 120 degrees of hip flexion, pain
 - » Indicates pain with hip flexion - complete local biomechanical exam. Proceed to the Active Seated Hip Rotation Tests.
- DN: Less than 120 degrees of hip flexion, no pain
 - » Indicates a hip mobility dysfunction for hip flexion - complete local biomechanical exam. Proceed to the Active Seated Hip Rotation Tests.
- DP: Less than 120 degrees of hip flexion, pain
 - » Indicates a painful hip flexion mobility dysfunction - complete local biomechanical exam. Proceed to the Active Seated Hip Rotation Tests.



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Test Objective for the Active Seated External Hip Rotation Test:

To test active external hip rotation in an unloaded lower-body posture with the hip flexed.

How to Perform the Active Seated External Hip Rotation Test:

Instruct the patient to assume a seated position, body in an upright and erect posture, and hands on hips (this allows you to see any hiking of the pelvis during testing). Have the patient externally rotate the hip as far as possible. Repeat the test on the other side.

Functional Criteria:

Normal external hip rotation is 40 degrees bilaterally.

Additional Information:

The patient should be able to complete the motion while keeping the knee flexed and the pelvis level. Watch for hip flexion versus external rotation.

Possible Findings:

- FN: 40 degrees or greater of active external hip rotation, no pain
 - » Consider hip external rotation normal. Proceed to the Active Seated Internal Hip Rotation Test.
- FP: 40 degrees or greater of active external hip rotation, pain
 - » Proceed to Passive Seated External Hip Rotation Test.
- DN: less than 40 degrees of active external hip rotation, no pain
 - » Proceed to Passive Seated External Hip Rotation Test.
- DP: less than 40 degrees of active external hip rotation, pain
 - » Proceed to Passive Seated External Hip Rotation Test.



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Test Objective for the Passive Seated External Hip Rotation Test:

To test passive external hip rotation in an unloaded lower-body posture with the hip flexed to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Seated External Hip Rotation Test:

Instruct the patient to assume a seated position, body in an upright and erect posture, and hands on hips (this allows you to see any hiking of the pelvis during testing). Passively externally rotate the hip as far as possible. Repeat the test on the other side if needed.

Functional Criteria:

Normal passive external hip rotation is 40 degrees bilaterally.

Additional Information:

The pelvis should remain level and the knee flexed while passively completing the motion.

Possible Findings:

- FN: 40 degrees or greater of passive hip external rotation, no pain
 - » Indicates an external hip rotation stability/motor control dysfunction with the hip flexed. Proceed to the Active Seated Internal Hip Rotation Test.
- FP: 40 degrees or greater of passive hip external rotation, pain
 - » Indicates pain in the hip for external hip rotation with the hip flexed - complete local biomechanical exam. Proceed to the Active Seated Internal Hip Rotation Test.
- DN: less than 40 degrees of passive hip external rotation, no pain
 - » Indicates there is a hip mobility dysfunction for external rotation with the hip flexed - complete local biomechanical exam. Proceed to the Active Seated Internal Hip Rotation Test.
- DP: less than 40 degrees of passive hip external rotation, pain
 - » Indicates painful hip mobility dysfunction for external rotation with the hip flexed - complete local biomechanical exam. Proceed to the Active Seated Internal Hip Rotation Test.



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Test Objective for the Active Seated Internal Hip Rotation Test:

To test active internal hip rotation in an unloaded lower-body posture with the hip flexed.

How to Perform the Active Seated Internal Hip Rotation Test:

Instruct the patient to assume a seated position, body in an upright and erect posture, and hands on hips (this allows you to see any hiking of the pelvis during testing). Have the patient internally rotate the hip as far as possible. Repeat the test on the other side.

Functional Criteria:

Normal internal hip rotation is 30 degrees bilaterally.

Additional Information:

The patient should be able to complete the motion while keeping the knee flexed and the pelvis level.

Possible Findings:

- FN: 30 degrees or greater of active internal hip rotation, no pain
 - » If no previous findings, consider this a weight-bearing ankle, knee and/or hip stability/motor control dysfunction and/or normal anatomical variances.
- FP: 30 degrees or greater of active internal hip rotation, pain
 - » Proceed to the Passive Seated Internal Hip Rotation Test.
- DN: Less than 30 degrees of active internal hip rotation, no pain
 - » Proceed to the Passive Seated Internal Hip Rotation Test.
- DP: Less than 30 degrees of active internal hip rotation, pain
 - » Proceed to the Passive Seated Internal Hip Rotation Test.



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Test Objective for the Passive Seated Hip Internal Rotation Test:

To test passive hip internal rotation in an unloaded lower-body posture with the hip flexed to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Seated Hip Internal Rotation Test:

Instruct the patient to assume a seated position, body in an upright and erect posture, and hands on hips (this allows you to see any hiking of the pelvis during testing). Passively internally rotate the hip as far as possible. Repeat the test on the other side if needed.

Functional Criteria:

Normal passive internal hip rotation is 30 degrees bilaterally.

Additional Information:

The pelvis should remain level and the knee flexed while passively completing the motion.

Possible Findings:

- FN: 30 degrees or greater of passive hip internal rotation, no pain
 - » Consider this a internal hip rotation with hip flexed stability/motor control dysfunction.
- FP: 30 degrees or greater of passive hip internal rotation, pain
 - » Indicates pain in the hip with internal hip rotation - complete local biomechanical exam.
- DN: Less than 30 degrees of passive internal hip rotation, no pain
 - » Indicates there is a hip mobility dysfunction with internal rotation with the hip flexed - complete local biomechanical exam.
- DP: Less than 30 degrees of passive internal hip rotation, pain
 - » Indicates painful hip mobility dysfunction for internal rotation with the hip flexed - complete local biomechanical exam.

